Exam.Code:0507
Sub. Code: 3891

2123

Master of Computer Applications
Third Semester
CS-2120: Interactive Computer Graphics

Time allowed: 3 Hours **Max. Marks: 80**

NOTE: Attempt five questions in all, including Question No. I which is compulsory and selecting one question from each Section.

x-x-x

I	a) What do you understand by aspect ratio? Discuss in brief.	(02)
	b) What is the purpose of a colour lookup table? Discuss in brief.	(02)
	c) What are the advantages of homogeneous coordinates? Discuss in brief.	(02)
	d) What do you understand by composite transformations? Discuss in brief.	(02)
	e) What is scissoring in context of computer graphics? Discuss in brief.	(02)
	f) What do you understand by vanishing point? Discuss in brief.	(02)
	g) What is the use of *initgraph* function? Discuss in brief.	(02)
	h) What is the difference between GLU and GLUT? Discuss in brief.	(02)
	SECTION A	
II	a) Discuss the advantages and disadvantages of raster scan displays compared to random scan displays. When would you choose one display over the other? Discuss.	(08)
	b) How does a shadow mask method function in a CRT display? What are the advantages and disadvantages of this method? Discuss.	(08)
III	a) Draw and explain the architecture of a raster-graphics system with a display processor, focusing on how the system functions.	(08)
	b) Differentiate between emissive and non-emissive displays. Also discuss the basic principles and working of LCD displays.	(08)
	SECTION B	
IV	a) Explain DDA line drawing algorithm and discuss its advantages and disadvantages.	(08)
	b) Derive the general 2D rotation matrix with a pivot point and explain the key distinctions between this matrix and the 2D rotation transformation about the origin.	(08)
V	a) Briefly discuss mid-point circle drawing algorithm and discuss its advantages and disadvantages.	(08)
	b) Prove that a uniform scaling and a rotation form a commutative pair of operations.	(08)
	SECTION C	
VI	a) Briefly discuss the uses and working principles of light pen and trackball.	(08)
	b) Discuss and differentiate between parallel and perspective projections.	(08)
VII	a) What are the essential functions to capture mouse input in computer graphics programming? Discuss the utility of mouse programming with the help of a suitable example. Appropriately assume any required information yourself.	(08)
	b) Discuss the use and working of Back-Face Detection method.	(08)
	SECTION D	
VIII	a) What are primitives in computer graphics? How do they relate to constants, actions, operators, and variables in graphics programming? Discuss.	(08)
	b) How can graphics images be saved and loaded from storage devices like disks? Discuss with the help of a suitable example.	(08)
IX	a) Explain with suitable example and code snippets, how animation can be used to visualize any sorting algorithm.	(08)
	b) Write a note on the important features of OpenGL.	(08)

x-x-x

Exam.Code:0507
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2123

Master of Computer Applications

Third Semester

CS-2121: Theory of Computation

Time allowed: 3 Hours **Max. Marks: 80**

NOTE: Attempt five questions in all, including Question No. 9 (Section-E) which is compulsory and selecting one question each from Section A-D.

x-x-x

SECTION A

1. (a) Minimize the following DFA, Let $M = (\{q_0, q_1, q_2, q_3, q_4\}, \{0,1\}, \delta, q_0, \{q_4\})$ be DFA where, $\delta(q_0, 0) = \{q_1\}, \delta(q_0, 1) = \{q_3\}, \delta(q_1, 0) = \{q_2\}, \delta(q_1, 1) = \{q_4\}, \delta(q_2, 0) = \{q_1\}, \delta(q_2, 1) = \{q_4\}, \delta(q_3, 0) = \{q_2\}, \delta(q_3, 1) = \{q_4\}, \delta(q_4, 0) = \{q_4\}, \delta(q_4, 1) = \{q_4\}$
(b) Discuss steps to construct Moore machine to find 2's complement of binary number.

(8+8=16)

2. (a) Differentiate between DFA, NFA and ϵ-NFA. Convert the following NFA to DFA:
Let $M = (\{q_0, q_1\}, \{0,1\}, \delta, q_0, \{q_1\})$ be NFA where,
$\delta(q_0, 0) = \{q_0, q_1\}, \ \delta(q_0, 1) = \{q_1\}, \ \delta(q_1, 0) = \phi, \ \delta(q_1, 1) = \{q_0, q_1\}$.
(b) Write short note on left and right linear grammars.

(8+8=16)

SECTION B

3. (a) List pumping lemma conditions for regular languages. Prove that $L = \{a^i b^i, i > 0 \mid\}$ is regular language or not?
(b) Discuss Chomsky normal form and Greibach normal form using examples.

(8+8=16)

4. (a) The CFG $(\{S, A, B, C\}, \{a, b\}, P, S)$, where P is $S \rightarrow AB|BC, \ A \rightarrow BA|a, \ B \rightarrow CC|b, \ C \rightarrow AB|a$. Using CYK, is $w \in L$ true or not? where $w = "baaba"$.
(b) Write short note on derivation tree and ambiguity with examples.

(8+8=16)

SECTION C

5. (a) Write short note on deterministic and non-deterministic push down automata.
(b) Construct a Turing machine that accepts the language $\{L = 0^n 1^n \mid n \geq 0\}$. Discuss reject state too.

(8+8=16)

6. (a) Construct a Turing machine to add two numbers.
(b) Write short note on universal Turing machine and Chomsky hierarchy of languages.

(8+8=16)

SECTION D

7. Write short notes on: compiler structure, top-down parser and compilation phases.

(6+6+4=16)

8. (a) Discuss in detail for pre and post conditions in compiler design.
(b) Discuss predictive parsers. Is backtracking related to predictive parsers? explain.

(8+8=16)

SECTION E (Compulsory Question)

9. (i) Design 1's complement using Moore machine.
(ii) List applications of finite automata.
(iii) List Turing machine 7-tuples.
(iv) How parsing is related to compilation?

$(4 \times 4 = 16)$

x-x-x

Exam.Code:0507
Sub. Code: 3893

2123

Master of Computer Applications

Third Semester

CS-2122: Advanced Java and Network Programming

Time allowed: 3 Hours Max. Marks: 80

***NOTE**: Attempt five questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I - IV.*

UNIT - I

I. Discuss the various child classes of the JComponent along with the methods. (16)

II. How the events of Frame and Dialog classes are handled in Java? (16)

UNIT - II

III. How text files are stored and retrieved (as records) in JDBC? Explain with example? (16)

IV. Discuss the following methods of the Naming class :-
lookup (), bind (), unbind (), rebind() and list () (16)

UNIT - III

V. WAP client server application in which the server can response to maximum 10 clients? On reaching maximum clients the server must respond to the client as "No More Slot"? (16)

VI. Write a Servlet Program that takes a Color name in an input box from the user and display the Web page in that Color? (16)

UNIT - IV

VII. What are features of EJB? Compare EJB and BDK? (16)

VIII. What are the various JSP Elements? Explain each in detail? (16)

UNIT - V

IX. Attempt the following:-

a) What are the container classes in Java Swing API?
b) What is Marshalling and Un-Marshalling?
c) Which method is invoked when the text of TextArea is changed?
d) What is Introspection?
e) What is Inet4Addressa and Inet6Addess?
f) List various server packages to run JSP?
g) What are the types of EJB?
h) What is socket? (8x2)

x-x-x

Exam.Code:0507
Sub. Code: 3894

2123

Master of Computer Applications

Third Semester

CS-2123: Mobile Communication and Application Development

Time allowed: 3 Hours Max. Marks: 80

***NOTE**: Attempt five questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I -IV.*

x-x-x

UNIT - I

I. Why 2G network is the most prominent mobile communication technology? Discuss the services and architecture of GSM in support of your answer. (16)

II. What do you understand by the concept of localization? Elaborate on various types of handover along with their impact on mobile communication. (16)

UNIT - II

III. Why is database hoarding a sensitive issue in mobile devices? Describe various data delivery mechanisms along with their advantages, limitations and suitable application examples. (16)

IV. Compare and contrast the following:-

a) Two popular languages for mobile application development

b) Two popular operating systems for mobile application development (16)

UNIT - III

V. Write note on

a) Android documentation

b) Android Emulator

c) Android Architecture

d) Application Context (16)

VI. Why is an activity the most pivotal element for application development? Discuss life cycle of an activity highlighting role of various call back methods with suitable examples. (16)

P.T.O.

UNIT - IV

VII. Describe various application resources along with steps involved in including the major ones while designing user friendly interface of a mobile application. (16)

VIII. Explain the role of android manifest file highlighting basic application settings and permissions for designing a mobile application. (16)

UNIT - V

IX. Attempt the following:-

a) Distinguish between Guided and Unguided transmission. (3)

b) Explain the concept of query processing in mobile database. (3)

c) Why are fragments used in Android application interface design? (3)

d) How is an activity registered in Android Application Development? (3)

e) What is the significance of radio interface in mobile communication? (2)

f) List major features of mobile devices popular in market. (2)

x-x-x